

FEBRIAN FITRYANIK SUSANTA

Geoinformatics | Geospatial Analytics | Human-Centered Transportation | Surveying | Geovisualization
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PROFILE

Geomatics lecturer, researcher, and Ph.D. student with experience in GIS, WebGIS, spatial data infrastructure, geospatial analytics, geovisual analytics, transportation accessibility, pedestrian safety, 3D spatial data, and geospatial community-service projects. Experienced in teaching, thesis supervision, technical training, peer-reviewed publication, and applied geospatial systems for public-sector and research partners. Current PhD researcher at the Department of Geomatics, National Cheng Kung University, Taiwan, develops a human-centered transportation framework by integrating data-driven geospatial intelligence and precision spatial surveying.

KEY SKILLS

- **Geospatial analytics:** GIS, spatial statistics, network analysis, spatial accessibility, human-centered transportation, urban analytics, and geovisualization.
- **WebGIS and spatial platforms:** LeafletJS, D3.js, OpenLayers, Mapbox, ArcGIS Online Apps, GeoNode, geoportals, and spatial information system development.
- **GIS, remote sensing, and data management:** QGIS Desktop, ArcGIS Desktop/Pro, Google Earth Engine, CloudCompare, Global Mapper, PostgreSQL/PostGIS.
- **Programming and analytics:** Python, JavaScript, C++, R, MATLAB, Tableau, DataPlotly, Microsoft Power BI, Microsoft Office, and Google Workspace.
- **Research, teaching, and delivery:** academic publishing, conference presentation, course design, stakeholder training, project-based learning, and community-service leadership.

PROFESSIONAL EXPERIENCE

Ph.D. Student / Research Assistant | Department of Geomatics, National Cheng Kung University | Feb 2024 - present | Tainan, Taiwan

- Conduct research on elderly pedestrian traffic safety, pedestrian space, sidewalk characteristics, and walkability for time-based route planning.
- Collaborate on spatial analysis, transport safety, high-definition mapping, and interpretable modeling studies with academic research teams.

Assistant Professor | Department of Geodetic Engineering, Universitas Gadjah Mada | Nov 2020 - present | Yogyakarta, Indonesia

- Design and deliver undergraduate and graduate-level courses in GIS, geomatics, spatial analysis, and related geospatial topics.
- Supervise theses and capstone projects while supporting student research, workshops, and academic publication activity.
- Lead and contribute to public-sector, research, and community-service projects involving spatial information systems and geospatial analysis.

Trainer in Basic GIS and WebGIS | Frasta Education Center, Center for Transportation and Logistics Studies UGM, and incidental programs | 2016 - 2024 | Indonesia

- Delivered GIS and WebGIS training for professional, student, and institutional audiences.
- Initiated project-based learning modules and organized participant workshops focused on applied geospatial tools and workflows.

Lecturer | Department of Geodetic Engineering, Universitas Gadjah Mada | Nov 2018 - Oct 2020 | Yogyakarta, Indonesia

- Taught core courses and supported curriculum delivery in geodetic and geomatic engineering.
- Organized student workshops on GIS tools and contributed to project-based classroom activities.

Researcher | Center for Spatial Data Infrastructure Development, UGM | Nov 2018 - Jan 2024 | Yogyakarta, Indonesia

- Contributed to national and regional research projects, technical reports, and prototype spatial data platforms.
- Presented findings and supported applied spatial data infrastructure work for government and research partners.

Teaching and Research Assistant | Department of Geodetic Engineering, Universitas Gadjah Mada | Before Nov 2018 | Yogyakarta, Indonesia

- Supported faculty in lectures, laboratory sessions, and faculty-led research activities.

SELECTED PROJECTS

- **Elderly Pedestrian Traffic Safety and Pedestrian Space:** sidewalk-characteristics research assistant for a Higher Education Sprout Project supported by NCKU and NSTC, Taiwan, 2024-present.
- **Study on Pilot Project for Upgrading 3D Spatial Data in Bogor, Indonesia:** mid-level researcher for the Ministry of Agrarian Affairs and Spatial Planning / National Land Agency, 2023.
- **Accessibility Analysis of Public Transportation in Yogyakarta City:** team leader, Department of Geodetic Engineering, Universitas Gadjah Mada, 2023.
- **Development of Spatial Information System for Yogyakarta City:** GIS expert for land and urban planning geoportal work, 2021-2023.
- **Land Database Development for Implementing 3D Cadastre and Pilot Project of 3D Cadastre for MRT Jakarta:** co-researcher, 2020-2021.
- **National Spatial Data Infrastructure review:** research assistant for the Geospatial Information Agency, Indonesia, 2020.

EDUCATION

Ph.D. Student | Department of Geomatics, National Cheng Kung University | Feb 2024 - present | Taiwan

- 18 credits completed; weighted average score: 93.4/100.

Master of Engineering in Geomatic Engineering | Universitas Gadjah Mada | Aug 2016 - Jun 2018 | Indonesia

- GPA: 3.93/4.00; 42 credits across 16 courses.
- Thesis: Network Analysis for Modeling Bus Public Transport Systems using Weight Parameters in Kulon Progo Regency.

Bachelor of Engineering in Geodetic Engineering | Universitas Gadjah Mada | Sep 2013 - Jul 2017 | Indonesia

- GPA: 3.87/4.00; 151 credits across 61 courses.
- Thesis: Visual Analytics of Spatio Temporal Data for Railway Network using Interactive Online Map.

Selected graduate coursework: Geodetic Surveying, Remote Sensing and Image Interpretation, Database Systems, Computer Graphics, GIS, Photogrammetry, Geospatial Analysis, Spatial Statistics, Applied GIS, Cartography, Web GIS, and Spatial Data Infrastructure.

SELECTED PUBLICATIONS

- Putra, I. G. B., Kuo, P. F., **Susanta, F. F.**, Tedjo, B. H., & Lord, D. (2026). Identifying Environmental Factors Related to Pedestrian Crash Frequency Across Age Groups. *Transport Policy*, 104235.
- Putra, I. G. B., Kuo, P. F., **Susanta, F. F.**, Tedjo, B. H., & Lord, D. (2026). GeoShapley-based interpretation of older adult pedestrian fatal vs injury crash frequency in dense urban environments. *Accident Analysis & Prevention*, 230, 108450.
- Tedjo, B. H., Kuo, P. F., **Susanta, F. F.**, & Putra, I. G. B. (2026). Identifying environmental factors related to motorcyclist crash rates: variable selection using spatial Random Forest with network distance and barriers. *Accident Analysis & Prevention*, 226, 108362.
- Susanta, F. F.**, & Aditya, T. (2020). Visualisasi Pemodelan Hasil Analisis Jaringan Angkutan Umum di Kabupaten Kulon Progo. *GEOMATIKA: Jurnal Badan Informasi Geospasial*, 26(1). <http://dx.doi.org/10.24895/JIG.2020.26-1.1085>.
- Aditya, T., Laksono, D., **Susanta, F. F.**, Istarno, Diyono, & Ariyanto, D. (2020). Visualization of 3D Survey Data for Strata Titles. *ISPRS International Journal of Geo-Information*, 9(5), 310. <https://doi.org/10.3390/ijgi9050310>.
- Susanta, F. F.**, Pratama, C., Aditya, T., Khomaini, A. F., & Abdillah, H. W. K. (2019). Geovisual Analytics of Spatio-Temporal Earthquake Data in Indonesia. *JGISE*, 2(2). <https://doi.org/10.22146/jgise.51131>.

SELECTED CONFERENCES

- Susanta, F. F.**, Kuo, P.-F., & Putra, I. G. B. (2026). Modeling dynamic walkability to support time-based route planning for older adults. XXV ISPRS Congress, Toronto, Canada.
- Ku, T. Y., **Susanta, F. F.**, Kuo, P. F., Shen, C. W., Lai, C. J., & Chen, N. Y. (2026). Integrating Multi-objective Optimization and Spatial Analysis for Site Selection and Improvement of Levee Crossing Facilities. WCTR 2026, Toulouse, France.
- Susanta, F. F.**, Kuo, P. F., & Putra, I. G. B. (2026). A GIS-Based Multi-Objective Optimization Framework for Age-Sensitive Pedestrian Route Choice. 2026 International Symposium on Remote Sensing, Matsue, Japan.
- Putra, I. G. B., **Susanta, F. F.**, Tedjo, B. H., & Kuo, P. F. (2025). Interpretable machine learning for crash severity analysis of food delivery motorcyclists. 46th Asian Conference on Remote Sensing, Makassar, Indonesia.
- Li, Y. T., Kuo, P. F., Chiang, K. W., Putra, I. G. B., & **Susanta, F. F.** (2025). Utilizing High-Definition Maps and Simulation Software to Enhance Autonomous Vehicle Safety. The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 48, 929-935.
- Susanta, F. F.**, Kuo, P. F., Putra, I. G. B., & Tedjo, B. H. (2025). Equitable Sidewalks for All: Enhancing Hospital Pedestrian Accessibility Through Data-Driven Assessment. 2025 International Symposium on Remote Sensing, Incheon, South Korea.
- Susanta, F. F.**, Kuo, P. F., & Putra, I. G. B. (2024). A Comparison of the Spatial Accessibility Model for Hierarchical Healthcare Facilities in Yogyakarta City, Indonesia. 29th International Symposium on Remote Sensing.
- Susanta, F. F.**, Muryanto, R., & Laksono, D. P. (2024). The 15-minute city of mass public transportation in Yogyakarta: case of Trans Jogja bus. Eighth Geoinformation Science Symposium 2023, SPIE.
- Aditya, T., Laksono, D. P., Atunggal, D., **Susanta, F. F.**, Widjajanti, N., Setiawan, M. B., Agam, N., & Wibisono, T. (2021). 3D Modelling, Validation and Visualization of 3D Parcels in First Registration for 3D Cadastre: Indonesia Case. 7th International FIG Workshop on 3D Cadastres.
- Susanta, F. F.** (2019). Visual analytics for railway network in Java Island. Sixth Geoinformation Science Symposium, SPIE.

SELECTED BOOKS

- Heliani, S., & **Susanta, F. F.** (2023). Teknologi dan Tata Kelola Produksi Garam: Aspek Geografis: Analisis Potensi Lokasi Produksi Garam. In A. A. Ivan (Ed.), *Teknologi dan Tata Kelola Produksi Garam* (pp. 1-25). Pustaka Belajar.
- Pribadi, O. D., Adianto, D., Amarona, M. Q., Mulyadi, D. S., Abimanyu, A., Dharna, C. S., & **Susanta, F. F.** (2021). Jala Citra I - 2021 Aurora Expedition. Hydro-Oceanography Centre Indonesian Navy.

HONORS AND AWARDS

- 2026 - Yuan Fu-Guo Scholarship Award, National Cheng Kung University.
- 2025 - Humanistic Street Hackathon, Ministry of the Interior, Land Management Agency, Taiwan: Excellent Team for Smart Embankment Crossing.
- 2024 - Veritas et Conscientia Scholarship Elite Scholarship, National Cheng Kung University.

SELECTED CERTIFICATES

- LiDAR Winter School, National Cheng Kung University, 2026.
- Lecturer Certification, Indonesian Ministry of Education, Culture, Research, and Technology, 2023.
- Professional Engineer Certification (Ir.), Institution of Engineers Indonesia, 2023.
- Public Awareness Program on Geographical Names and Toponyms, Geospatial Information Agency Indonesia, 2023.
- Networking Development Training, Universitas Gadjah Mada, 2021.
- IELTS Advanced, REAL - IONS International Education, 2021.
- ESRI MOOCs: Spatial Data Science, The Location Advantage, Going Places with Spatial Analysis, and Cartography, 2019-2020.
- Geographic Information System Training, ESRI, 2020.
- Training of Trainers Faculty Advisor for the Student Creativity Program, 2019.
- Workshops on Academic Writing and Publishing, Nature Research Academy and Enago Academy, 2019.

TEACHING, TRAINING AND MATERIALS

- Courses and training areas include GIS, WebGIS, geospatial analysis, transportation accessibility, spatial data infrastructure, cartography, photogrammetry, remote sensing, database systems, and computer graphics.
- Developed and delivered practical, project-based geospatial training for university, government, professional, and community-service audiences.
- Supervised theses and capstone work and supported student workshops on applied GIS tools and spatial data workflows.

LANGUAGES

Indonesian: native | English: professional working proficiency.

ASSOCIATIONS AND ONLINE PROFILE

Indonesian Surveyor Association (ISI): <http://www.isi.or.id/>
Institution of Engineers Indonesia (PII): <http://www.pii.or.id/>
ISPRS Student Consortium (ISPRS SC): <https://sc.isprs.org/>